

SMOKING CESSATION

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I. INTRODUCTION

Importance

Smoking is a significant factor for many chronic diseases such as coronary artery disease, lung disease, cerebrovascular disease, cancers, and complications during pregnancy. On average, a smoker dies 14 years prematurely due to tobacco-related diseases compared to a non-smoker. Tobacco use is a chronic condition that often requires repeated cessation interventions in different forms. Hence, smoking cessation is relevant in the management of many chronic diseases.

Epidemiology

According to the MOH National Population Health Survey, the prevalence of daily smoking was 10.6% in 2019 and 9.2% in 2022.

II. SCREENING AND DIAGNOSIS

Initial Assessment

Assessment of smoking habits (estimated sticks/day) should be done once-off for all patients unless there is a change in smoking habit. Smokers with chronic disease should have a smoking assessment done annually and smoking cessation counselling should be provided.

III. MANAGEMENT AND FOLLOW-UP

Initial Management

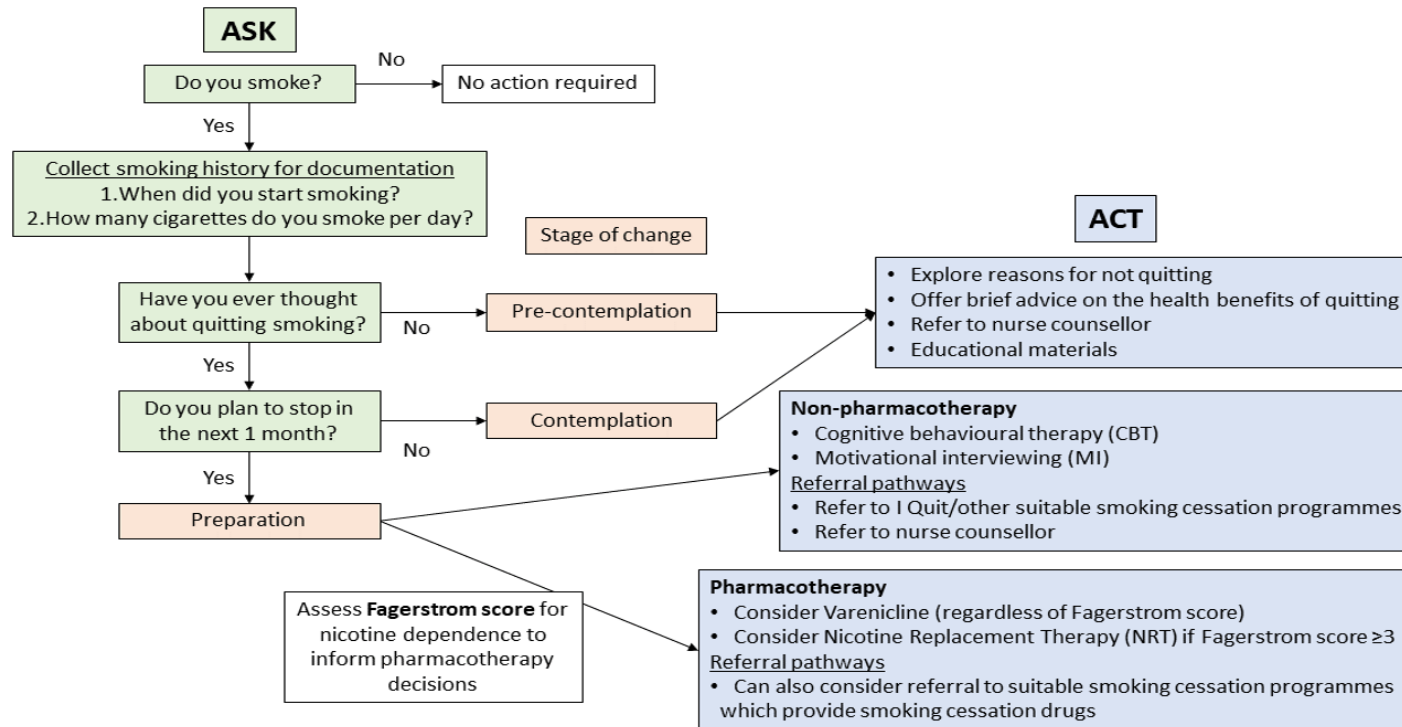
For identified smokers, tobacco use and dependence treatment interventions can be carried out at 3 levels:

- Level 1: The 2As or the ABC Approach (Basic Level Intervention) – **[recommended for NUP doctors]**
- Level 2: The 5As Approach (Intermediate Level Intervention)
- Level 3: Intensive Behavioural Support (Advanced Level Intervention) – not covered in this CPG

Level 1: The 2As Approach (Recommended for NUP doctors)

The 2As approach provides a helpful framework to guide healthcare providers in smoking cessation

1. **Ask** about smoking
2. **Act** to help all tobacco users quit



Licensing for Bupropion for smoking cessation under the Zyban® brand has been discontinued. The existing brand of Bupropion, Wellbutrin®, is only approved for treatment of major depressive disorder.

Level 1: The ABC Approach (Recommended for NUP doctors)

The aim of the ABC approach is to incorporate tobacco use cessation advice as a routine opportunistic first-line intervention when providing consultation to patients.

1. **ASK and document status of tobacco use for every patient:** *Do you smoke? How many sticks? Do you want to quit?*
2. **BRIEF ADVICE** to stop smoking to every smoker, **regardless of intention to quit:**
 - a) Risk of smoking
 - b) Health benefits of quitting (refer to [Appendix 1](#))
 - c) Relationship between smoking and current medical condition, the risk of continuing with the habit and benefit of quitting
3. **CESSATION** support for every patient who wants to quit:

Patient who wants to quit:

- a) Refer to specialised smoking cessation services (**NUP pharmacy-led Smoking Cessation Clinics**, HPB Quitline 1800 438 2000 **or the I Quit programme** <https://www.healthhub.sg/programmes/iQuit>)
- b) Assess for nicotine dependence and KIV start pharmacotherapy (refer to [Appendix 2](#)) – **Consider referral to smoking cessation clinic**

Patient not keen to quit:

- a) Ask for reasons for not wanting to quit now
- b) Ask for reasons which may make them quit in future
- c) Provide information for smoking cessation service
 - E.g., I Quit page on HealthHub

Level 2: The 5As Approach

The 5As approach is recommended for health professionals (doctors, nurse educators, care managers) who can spend about 15-20 minutes to initiate assessment and cessation interventions with patients who use tobacco.

1. **Ask** and document status of tobacco use for every patient
2. **Advise to quit**: provide brief, firm, personalised advice to trigger a quit attempt and document the advice given
3. **Assess willingness to quit**: consider using stages of change model (see below)

Stages of Change		Assisting the Smokers
Pre-Contemplation	Not motivated to stop smoking	Provide motivational intervention (see section on Non-Pharmacological Interventions)
Contemplation	Motivated to stop smoking but has not set a quit date	Refer for smoking cessation services
Action	Motivated to stop smoking within 1 month, <u>or</u> Has already stopped for < 1 month	Refer for smoking cessation services
Maintenance	Has stopped smoking for ≥ 1 month	Provide brief effective relapse prevention advice

4. **Assist** in quit attempt
5. **Arrange follow up** to check on patient's progress in their attempts to quit

Non-Pharmacological Interventions

Behavioural Support

1. Indicated for all smokers who are keen to quit
2. Can be done by any health professionals (physician, nurses, pharmacists, psychologists, and trained advisors). **Success rate** is the same regardless of the profession of the advisors
3. Individual basis or in a group or telephone support
4. When possible, need multiple intensive behavioural support sessions with each session lasting > 10 minutes
5. Should provide patients with problem and coping skills, and intra treatment social support and encouragement

Types of behavioural interventions:

1. Cognitive Behavioural Therapy

a) ***Cognitive / experiential strategies:*** Focus on retraining the way a patient thinks

<i>Consciousness Raising</i>	• <i>Educate the patient on the negative health effects of smoking</i>
<i>Emotional Arousal</i>	• <i>Encourage the patient to experience and express their feelings about smoking</i> • <i>Explore positive benefits of a smoke-free lifestyle</i>
<i>Environmental Re-Evaluation</i>	• <i>Encourage the patient to consider the harmful effects of smoking on the environment, e.g., how second-hand smoke affects family members</i>
<i>Social Liberation</i>	• <i>Discuss how the society is making it easier for non-smokers (e.g., smoke free public spaces, heavier tax on cigarettes)</i>
<i>Self-Re-Evaluation</i>	• <i>Explore what being a smoker means to the patient, how it contradicts with his/her desires for better health, and what being a non-smoker means to the patient</i>

b) **Behavioural strategies:** *Specific actions to attain and maintain freedom from smoking*

Self-Liberation	• <i>When the patient is ready, encourage patient to quit smoking</i> • <i>Set a quit date or develop a quit plan</i> • <i>Encourage positive 'self-talk'</i>
Helping Relationships	• <i>Assist the patient in identifying a smoking quit-buddy</i> • <i>Arrange for regular follow-ups</i>
Reinforce Management	• <i>Praise the patient</i> • <i>Encourage the patient to set up a reward system: positive 'self-talk', pleasant activities, or special gift / trip using the money saved on cigarettes</i>
Stimulus Control	• <i>Encourage the patient to throw all smoking paraphernalia on the quit date</i> • <i>Avoid fellow smokers</i> • <i>Change routines: identify alternative plans for routines associated with smoking</i> ○ <i>Morning cigarettes: e.g., take a shower and brush teeth first thing in the morning, take a brisk walk, having breakfast at smoke-free eateries</i> ○ <i>While driving: remove all tobacco from car, re-model car interior, use oral substitutes (gum, sugarless candy), listen to the radio or music</i> ○ <i>While on the phone: stand while talking, limit call duration, change phone location, keep hands occupied by doodling or sketching</i> ○ <i>After meals: get up and do dishes, brisk walking, call friends</i>
Counter Conditioning	• <i>Relaxation techniques: deep breathing, positive imagery, and muscle relaxation → smoking craving will pass within 5-10 minutes</i> • <i>Distractive thinking</i> • <i>Abstinence from alcohol and other drugs</i> • <i>Oral substitutes: gum, sugarless candy, straws, toothpick</i> • <i>Exercise (not dieting): KIV pharmacotherapy to delay weight gain (NRT, bupropion)</i>

2. Motivational Interviewing

Some key ideas of motivational interviewing include:

- a) Express empathy
- b) Listen out for discrepancy between current behaviour and long term goals
- c) Roll with resistance
- d) Active listening (“OARS”)
 - i. Ask **O**pen-ended question – e.g., “What do you think about stopping smoking?”
 - ii. Use **A**ffirmation – e.g., “You did well in stopping smoking previously.”
 - iii. Use **R**eflection – e.g., “So you feel that on one hand smoking relieves stress but on the other, it harms your health.”
 - iv. **S**ummarise – e.g., “We have discussed these issues, how should we move on from here?”

Pharmacological Interventions

The Fagerstrom Test (see [Appendix 3](#)) for Nicotine Dependence **can be used** to determine **the** level of dependence on nicotine. Pharmacotherapy is generally continued for at least 3 months **and should be given in combination with counselling for best results.**

First line pharmacotherapy options **include the following** (refer to [Appendix 2](#) for algorithm):

Nicotine Replacement Therapy (Available in NUP)

Subsidised nicotine replacement therapy (NRT) has been made available at public healthcare institutions since February 2023, and can be given at the pharmacy-led smoking cessation clinics in NUP.

a) How it works

- Nicotine is the main substance in tobacco that causes addiction, **however it is the** other chemicals in combusted tobacco which **are detrimental to health**
- NRT therefore works by reduces craving and withdrawal symptoms by providing some of the nicotine that is normally obtained from cigarettes, without providing the harmful components of tobacco smoke

b) Availability and efficacy

- Available as patches, gums, and lozenges
- All forms at equivalent dose are similarly effective in aiding long term smoking cessation by increasing quit rate by 50 - 70%

c) Combination NRT [patch + oral form (gum or lozenge)] is more efficacious than NRT monotherapy

- Patch provides a steady background nicotine level
- Lozenge or gum provides relief for breakthrough craving as needed
- Consider combination NRT if smoker is unable to quit using one NRT product alone, or experiences cravings using only one form of NRT

d) Pre-cessation nicotine patches (21 mg/24 hr patches OR 25 mg/16 hr patches)

- Can be given TWO weeks prior to smoking cessation to help smokers prepare for **the quit attempt**
- **No change in the use of NRT during the actual** quit attempt

e) Reduce to quit (gradual quitting)

- Recommended approach involves smokers using NRT to reduce the number of cigarettes they smoke before stopping completely within 6 months

A meta-analysis found that **neither reduction-to-quit nor abrupt quitting interventions resulted in superior long-term quit rates when compared with one another.**

f) Safety

- Using NRT is always safer than continuing smoking
- Can be used by people with stable CVD
 - Use with caution in people with recent MI, unstable angina, severe arrhythmia, and recent cerebrovascular events
- Can be considered for use in smokers aged 12 - 17 years
- Breastfeeding smokers
 - Nicotine passes from the mother to child through breast milk, depending on the nicotine concentration in the maternal blood
 - Nicotine in breast milk is unlikely to be dangerous
 - Encourage women who smoke to continue breastfeeding and provide them with strategies to minimise the potential harm to their child associated with second-hand smoke

Availability of NRTs in NUP (as of October 2023)

More details on NRT use under section on Smoking Cessation Clinics

NRT	Drug Class	Price	Remarks
nicoRETTE 2mg GUM 30s	Non-standard	\$8.70	
nicoRETTE <4MG> GUM 30s	Non-standard	\$8.70	Not available in some branches
nicoTINELL 1mg LOZENGES 36s	Non-standard	\$10.44	
nicoTINELL TTS 10 7s	Non-standard	\$12.53	
nicoTINELL TTS 20 7s	Non-standard	\$12.53	
nicoTINELL TTS 30 7s	Non-standard	\$12.53	

* All NRTs are non-standard but Medisave-claimable if patient has any of the following CDMP conditions and diagnosis code(s) which are added into the Epic encounter: 1) DM and Pre-diabetes, 2) Hypertension, 3) Lipid disorders, 4) Asthma, 5) COPD, 6) Chronic kidney disease, 7) Stroke, 8) Ischemic heart disease

* From 1 Feb 2023, MAF subsidies for NRTs are available for smokers who meet clinical MAF criteria – the patient needs to:

- Have a documented Fagerstrom score of 3 and above
- Be enrolled in NUP's smoking cessation programme and this should be documented in the clinical notes
- Attend a minimum of 3 smoking cessation programme sessions, with each session recommended to be longer than 10 minutes

1. Varenicline (Not Available in NUP)

a) How it works (nicotinic receptor partial agonist)

- Acts at ACh receptor in the reward centre in the brain **by binding** with high affinity to $\alpha 4\beta 2$ nicotinic ACh receptor, where it acts as a partial agonist to alleviate symptoms of craving and withdrawal
- If a cigarette is smoked, **varenicline** prevents inhaled nicotine from activating the $\alpha 4\beta 2$ receptor sufficiently to cause the pleasure and reward response

b) Efficacy

- Can increase the chances of successful smoking cessation by 2 - 3 fold compared to pharmacologically unassisted smoking cessation attempts

- Longer term use (a second 12-week course) reduces relapse for up to one year in people who have successfully quit at the end of week 12
- Limited evidence on the use of varenicline in combination with other therapies

c) Safety

- Patient with current or past history of mental illness
 - Safe and effective in smokers with schizophrenia, or **stably treated current or past** depression
 - EAGLES trial (Anthenelli 2016):
 - Nil significant increase in the rates of moderate-to-severe neuropsychiatric adverse events in those taking varenicline compared to those using placebo, bupropion or nicotine patch, in smokers with or without mental illness
 - Those with mental illness in all treatment groups had higher rates of neuropsychiatric adverse events than those without mental illness.
- Smokers using varenicline should report unusual mood changes, depression, behaviour disturbance and suicidal thoughts
→ stop using medicine if these occur

2. Bupropion Sustained Release **(Not Available in NUP)**

a) How it works

- Originally developed as an antidepressant
- Reduces the urge to smoke and symptoms from nicotine withdrawal
- The quit attempt is generally initiated a week after starting pharmacotherapy

b) Efficacy

- Effective in patient with depression, cardiac disease, respiratory disease (including COPD)
- Less effective than varenicline, same efficacy as NRT monotherapy
- Limited evidence on the efficacy of combining bupropion with NRT or with varenicline

Summary on effectiveness of treatment from a Cochrane review:

1. NRT, bupropion and varenicline all improve the chances of quitting, with a low risk of harm.
2. NRT and bupropion have similar efficacy: **both increased quit rates by 80% as compared to placebo**
3. Varenicline is more effective than NRT monotherapy or bupropion
4. Varenicline more than doubled the chances of quitting compared with placebo
5. Combination use of NRT is more effective than NRT monotherapy
6. Combination NRT is as effective as using varenicline

Role of Non-Doctor Team Members

Availability of Smoking Cessation Clinics (SCC) in NUP (Run by Pharmacists)

- Cost (as of October 2023)

	Singapore Citizen Adult / Elderly / Child	PR	NR	Remarks
SCC	\$10.50	\$21	\$30	- Follow-up sessions which require counselling will be charged the same amount - FOC for top up of NRT or for short sessions within 12 weeks of first session or follow-up phone call

- Clinicians can refer suitable patients to SCC by ordering “TCU Pharm Smoking Cessation (SCC)” in Epic.
- The first SCC visit will be an in-person consult. After completion of 1st session, SCC pharmacist to offer 2 subsequent follow-up sessions (recommended at 4th week and 12th week) either in person or by phone consult.
- Where necessary, SCC pharmacist will draft NRTs as appropriate for the Dr to co-sign the prescription (individual clinics have their own workflow)

DRUG		SUGGESTED DOSAGE	ADVERSE EFFECTS	DIRECTIONS OF USE
NICOTINE REPLACEMENT THERAPIES (NRT)	Nicotinell® patch (NS) Around the clock relief from cravings <i>Available in 3 strengths:</i> - 21 mg/24 hr - 14 mg/24hr - 7 mg/24hr \$\$\$\$	a) Light smokers < 20 cig/day: Start with 14 mg/24 hr patch x 8 weeks, then 7 mg/24 hr patch x 4 weeks b) Heavy smokers > 20 cig/day: Start with 21 mg/24 hr patch x 4 weeks, then 14 mg/24 hr patch x 4 weeks, then 7 mg/24 hr patch x 4 weeks	ADVERSE EFFECTS: - Skin irritation / rash at application site - Insomnia & abnormal dreams (especially with 24 hr patches) - Headache - Nausea - Dizziness - Gastrointestinal complaints - Palpitations	- Apply a patch onto clean, dry, non-hairy skin, and press firmly For Nicotinell 24 hr patch: - Apply in the morning upon rising. - Re-apply every 24 hours. For Nicorette 16 hr patch: - Apply in the morning upon rising, and remove at bedtime. - Re-apply the next morning

DRUG		SUGGESTED DOSAGE	ADVERSE EFFECTS	DIRECTIONS OF USE
	Nicorette® patch (Not available in NUP) Around the clock relief from cravings <i>Available in 3 strengths:</i> - 25 mg/16 hr - 15 mg/16 hr - 10 mg/16 hr	a) Light smokers < 15 cig/day: Start with 15 mg/16 hr patch x 8 weeks, then 10 mg/16 hr x 4 weeks b) Heavy smokers > 15 cig/day: Start with 25 mg/16 hr x 8 weeks, then 15 mg/16 hr x 2 weeks, then 10 mg/16 hr x 2 weeks	CONTRAINDICATIONS: - Non smokers - Children < 12 years - Hypersensitivity to nicotine or any component of the patch - Disease of the skin that may complicate the patch therapy - Recent heart attack or stroke, unstable angina, and severe cardiac arrhythmia	- Remove used patch before applying a new one. - Alternate application site to minimize skin irritation. - Application sites: hip, chest, upper arm / thigh - Do not use more than 1 patch at a time
	Nicotinell® gum On the go relief during breakthrough cravings <i>Available as 2 mg and 4 mg gums</i> \$\$\$\$	Chew on 1 piece when the urge to smoke starts a) Light smokers: < 20 cig/day or 1st cig > 30 mins after waking: Use 2 mg gum 8 – 12x/day b) Moderate smokers: 20-30 cig/day: Low dependency: 2 mg 8 – 12x/day High dependency: 4 mg 8 – 12x/day c) Heavy smokers: 30 cig/day or 1st cig < 30 mins after waking: Use 4 mg gum 8 – 12x/day 2mg gums max 25/day 4mg gums max 15/day - Tx should continue for at least 3 months - Taper the dose slowly after 3 months - Discontinue tx once dose of gum has been reduced to 1-2 pieces per day	ADVERSE EFFECTS - Jaw fatigue - Mouth and throat soreness - Headache - Hiccups - Nausea - Stomach discomfort - Cough and irritation of the throat - Increased salivation CONTRAINDICATIONS - Non smokers - Children < 12 years old - Hypersensitivity or any component of the gum - Recent heart attack or stroke, unstable angina, and severe cardiac arrhythmia	- Chew intermittently until a peppery taste becomes strong - Park it in between cheek and gum for nicotine to be absorbed. - Chew again when the taste has faded (about 1 minute later). - Repeat the routine for 30 minutes. - Chew slowly to avoid hiccups and gastrointestinal side-effects. - Avoid acidic food (e.g., citrus fruits, soft drinks, coffee, juices) 15 minutes before chewing the gum - Not suitable for use in patients with dentures.

DRUG		SUGGESTED DOSAGE	ADVERSE EFFECTS	DIRECTIONS OF USE
	Nicorette® gum (Not available in NUP) On the go relief during breakthrough cravings <i>Available as 2mg and 4mg gums</i>	a) Light smokers (< 20cig/day or 1st cig> 30mins after waking): Use 2 mg gums b) Heavy smokers (>20 cig/day or 1st cig < 30 mins after waking): Use 4mg gum • Full duration 16 weeks • Use 8-12 pieces /day for 12 weeks, then • Use 4-6 pieces/day for the next 2 weeks, then • Use 1-3 pieces/day • Gradually reduce to zero		
	Nicotine lozenge (Nicotinell®) On the go <u>discreet</u> relief during breakthrough cravings <i>Available as 1mg and 2mg lozenges</i> <i>(Only 1 mg lozenges available in NUP)</i> \$\$\$\$	a) Light smokers (< 20 cigs/day): Use 1mg lozenge b) Moderate smokers (20 – 30 cigs/day): Use 1 mg or 2 mg lozenges c) Heavy smokers (> 30 cigs/day): use 2 mg lozenges • Suck on 1 piece every 1 – 2 hours. • Usual dose: 8 – 12 lozenges/day, max 25 lozenges/day. • Start to taper as tolerated after 12 weeks. • Stop using upon reducing to 1 – 2 lozenges/day. • Minimum therapy is 8 weeks • Do not use for more than 6 months	ADVERSE EFFECTS • Throat irritation • Coughing • Hiccups • Stomach discomfort • Dizziness • Headache • Increased salivation CONTRAINDICATIONS • Phenylketonuria • Non-tobacco users • Children< 12 years old • Hypersensitivity or any component of the lozenge • Recent heart attack or stroke, unstable angina, and severe cardiac arrhythmia	• Suck slowly until peppery taste becomes strong • Park it in between cheek and gum for nicotine to be absorbed. • Suck again when the taste has faded. • Repeat sucking routine for 30 minutes. • Do not chew or swallow the lozenge. • Avoid acidic food (e.g., citrus fruits, soft drinks, coffee, juices) 15 minutes before sucking the lozenges

DRUG		SUGGESTED DOSAGE	ADVERSE EFFECTS	DIRECTIONS OF USE
NON-NRT	Bupropion SR (Not available in NUP)	Days 1 – 3: 150 mg OD Days 4 – week 12: 150 mg BD - Consider lower dose of 150 mg OD in patient who cannot tolerate side effects of higher dose - Start 1 – 2 weeks before target quit date since it takes five to seven days to reach steady-state blood levels. - If nil significant progress is seen by week 7 → unlikely to succeed → consider discontinuing treatment - Maintenance therapy: up to 6 months	ADVERSE EFFECTS - Seizures (0.1%): most important adverse effect - Common adverse effects: insomnia, headache, dry mouth, nausea, dizziness, and constipation - Neuropsychiatric events (e.g., behavioural change, mood change, suicidal thoughts, attempted suicide) during treatment or treatment discontinuation CONTRAINDICATIONS - Children < 18 years old - Current or past hx of seizures - Past hx of head trauma - CNS tumour - Alcohol abuse - Current or past hx of eating disorders - Patient taking MAO inhibitors - Use with caution in patient taking medicines that lower seizure threshold (antidepressants, anti-malarials, and OHGAs), elderly, diabetics, patient with hepatic or renal impairment, pregnant and breastfeeding women	- Closely monitor for clinical worsening, suicidality or unusual changes in behaviour, particularly during first 1 – 2 months of therapy - When combined with NRT, monitor BP
	Varenicline (Champix®) (Not available in NUP) <i>Available as 0.5 mg and 1 mg tabs</i>	Days 1 – 3: 0.5 mg OD Days 4 – 7: 0.5 mg BD Day 8 onwards: 1 mg BD until the end of 12 weeks course Optional week 12 onwards: 1 mg BD until the end of additional 12 weeks course (total 24 weeks course)	ADVERSE EFFECTS - Nausea*(dose-dependent): most common (30% of users) - Abnormal dreams - Headache - Insomnia	- Take medicine whole with water and food to help reduce nausea - Closely monitor for clinical worsening, suicidality or unusual changes in behaviour, particularly during first 1 – 2 months of therapy.

DRUG	SUGGESTED DOSAGE	ADVERSE EFFECTS	DIRECTIONS OF USE
	• Consider lower dose in: ○ Patient who cannot tolerate side effects ○ Renal dose (Cr clearance < 30 mL/min): ➢ Daily dosage is 1 mg/day ➢ Start with 0.5 mg/day for 3 days then increasing to 1 mg/day ○ Nil dose adjustment needed when used in elderly or patient with hepatic impairment • Start 1 – 2 weeks before target quit date • Patient who are not ready to quit at the 2nd week should continue using varenicline for several more weeks → may quit between day 8 – 35 of varenicline treatment • Consider lower dose in patients unable to tolerate side effects • Minimum treatment duration: 12 weeks • Consider longer use of additional 12 weeks to maintain success for patient who successfully quit smoking	• Neuropsychiatric events (e.g. behavioural change, mood change, suicidal thoughts, attempted suicide) CONTRAINDICATIONS • Children <18 years old • ESRF (excreted almost entirely in kidney) • Use with caution in patient with hx of psychiatric illness, pregnant and breastfeeding women, elderly, and patient with renal impairment	

^ Cost \$ <\$10 per month \$\$ \$10 - <\$20 per month \$\$\$ \$20 - <\$30 per month
 \$\$\$\$ \$30 - <\$60 per month \$\$\$\$\$ \$60 - <\$90 per month \$\$\$\$\$\$ \$90 or more per month
 Amount payable depends on patient MAF status
 # Average monthly cost, assuming usage of 1 box/week

IV. SPECIAL SITUATIONS

	Recommended 1st line pharmacotherapy
Psychiatric illness	• Mild untreated depression: bupropion*, combination NRT, varenicline • Severe depression: refer to psychiatrist • Bipolar disorder: NRT, varenicline (avoid bupropion) • Schizophrenia: bupropion, varenicline (limited data on NRT)
Cardiovascular disease	• Stable CVD: same as general population- bupropion, NRT, varenicline
Seizures	• Safe options: varenicline and NRT (bupropion is contraindicated)
Preoperative smokers	• Effective: varenicline, NRT
Pregnant smokers	• First line: Non-pharmacological smoking cessation counselling • For those unable to quit or at risk of continued smoking (> 10 cigs/day, smoking later in pregnancy, unsuccessful past quit attempts): consider add-on NRT or bupropion (avoid varenicline) ○ NRT appears safe but its efficacy in pregnancy is not well-established ○ Bupropion is efficacious but is associated with a small increased risk of congenital malformation (cardiac abnormalities)
Light smokers < 10 cigarettes per day	• Lower dose of NRT, standard dose bupropion, standard dose varenicline
Smokers willing to reduce but not ready to quit in 1 month	• NRT, varenicline
Smokers concerned about post cessation weight gain	• Bupropion

**: preferred option*

V. ABBREVIATIONS

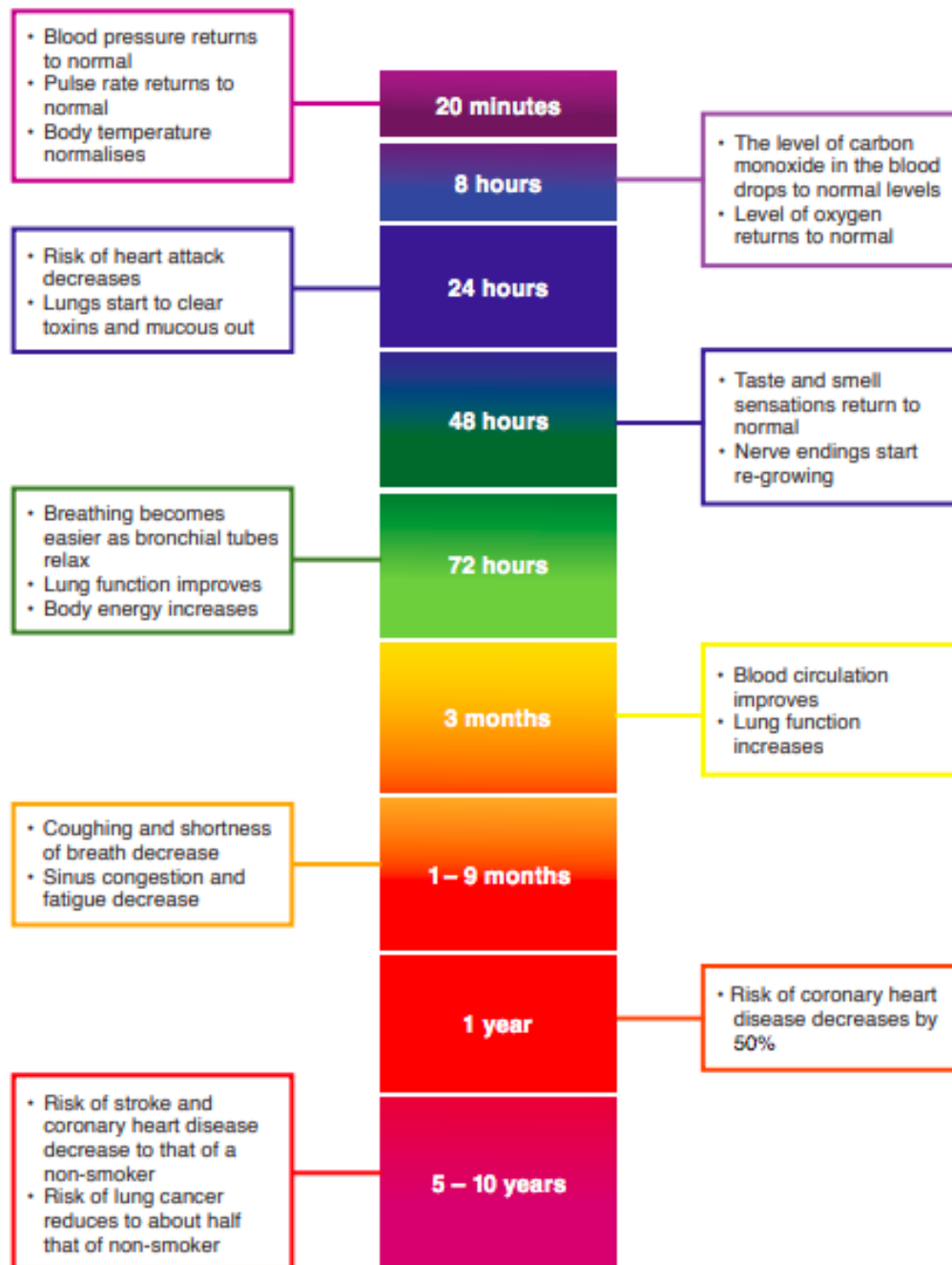
NRT Nicotine Replacement Therapy
SCC Smoking Cessation Clinics

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APPENDIX

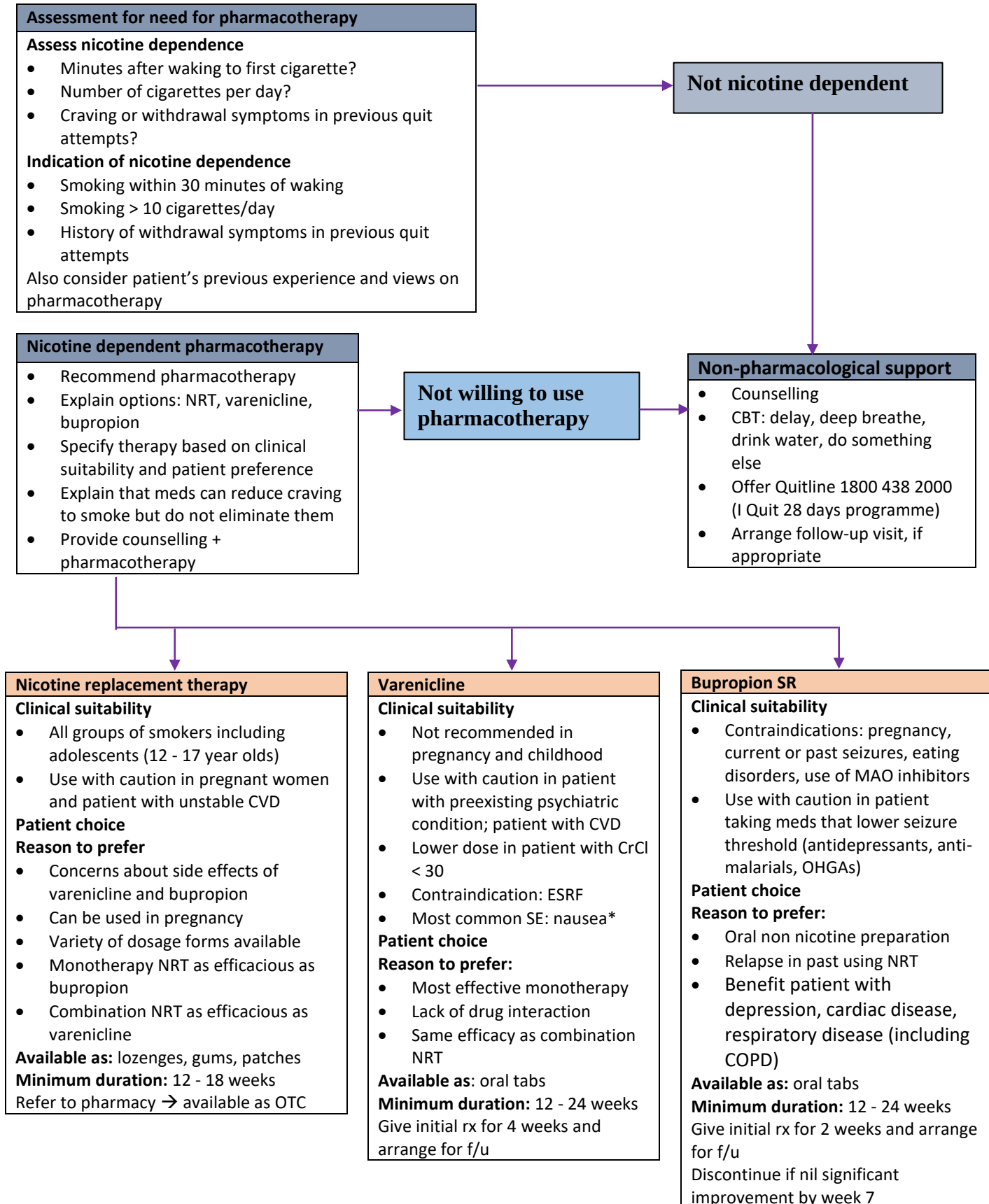
Appendix 1 – Health Benefits of Quitting Tobacco Use



Adapted from Health Promotion Board: http://www.hpb.gov.sg/hpb/default.asp?pg_id=2564

Appendix 2 – Pharmacotherapy Treatment Algorithm

(Adapted from RACGP Supporting Smoking Cessation: A Guide for Health Professionals)



Appendix 3 – The Fagerstrom Test for Nicotine Dependence

(adapted from the American Association of Respiratory Care)

How soon after waking up do you smoke your first cigarette?	<5 minutes 5–30 minutes 31–60 minutes >60 minutes	☐ 3 ☐ 2 ☐ 1 ☐ 0
Do you find it difficult to refrain from smoking in places where it is prohibited? E.g. parks, hospital compounds	Yes No	☐ 1 ☐ 0
Which cigarette would you hate to give up?	The first in the morning Any other	☐ 1 ☐ 0
How many cigarettes a day do you smoke?	≥31 21–30 11–20 ≤10	☐ 3 ☐ 2 ☐ 1 ☐ 0
Do you smoke more frequently in the morning?	Yes No	☐ 1 ☐ 0
Do you smoke even if you are sick in bed most of the day?	Yes No	☐ 1 ☐ 0
Total score		
Score	0–2: Very low dependence	5–7: Moderate-high dependence
	3–4: Low dependence	8–10: Very high dependence